

Exact Path Representations for Finite-Catalog Renewal Design

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Abstract. A renewal provider chooses a paid, limited catalog of terminal commands and allocates an accepted obligation across heterogeneous histories. We derive exact selected-boundary resource and price representations that preserve individual expected caps, realization ceilings, and the original promise. History-level component packing extends the representations to heterogeneous increasing concave terminal rewards. The rational decision problem is NP-complete even with unit linear reward, zero service costs, uniform probabilities, and a nonbinding command budget. In contrast, common expected caps admit an exact two-command reduction under a common reward. For rational quadratic primitives, a price oracle is quadratic in catalog size and linear in history count and command budget. It supports grid-free finite exact branching and independently verified intervals, including interrupted runs. A polynomial additive resource scheme remains available, with cubic catalog and quadratic budget dependence after substituting inverse normalized accuracy. Forced-command price bounds also permit safe exclusion of possible anchors. Matched-limit synthetic experiments vary history count, catalog size, budget, accuracy, and charges; exact reduction challenges expose unsuccessful cases as well as favorable performance. The contribution is the structural reduction, complexity boundary, and original-instance certification, not an empirical claim of universal solver superiority.

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1 Introduction

A renewal provider has accepted a total obligation but must implement it through a limited catalog of terminal commands. Different histories have different expected caps, realization ceilings, and costs of intermediate service. The provider chooses both the installed commands and the allocation of the obligation across histories. Opening charges discourage unnecessary commands. A useful algorithm must optimize these decisions jointly: changing the obligation or adding commands to repair an approximation changes the problem the provider agreed to solve.

This paper identifies two exact path representations of that joint problem. The first is a primal selected-boundary representation for a common increasing concave terminal reward. Every neighboring pair of selected commands owns a terminal-reward layer and the service costs of the histories whose eligibility ends at that boundary. These layers admit an explicit history-level reconstruction. Their capacities sum to the original available obligation, regardless of how the command book partitions histories. A uniform resource approximation can consequently be repaired at the original promise and command budget.

A history-level component-packing argument extends the primal representation to heterogeneous rewards by replacing a common chord with a concave within-edge allocation profile. The second representation is an exact priced path. It permits different increasing concave terminal rewards across histories. At a fixed obligation price, the optimal response of each history separates into its positive chord increments and its terminal service support. These terms can again be assigned to selected boundaries. One ordered dynamic program then optimizes all feasible anchors and command books. It computes a genuine original-space Lagrangian bound, not a bound on a gridded or relaxed-feasibility surrogate. The priced identity does not require different histories to share an ordering of marginal rewards; each history only needs its own concave ordering.

We first establish a complexity frontier. The exact design decision problem is NP-complete even with common unit linear reward, zero service costs, uniform history weights and a nonbinding command budget. A direct subset-sum construction gives the proof; the obstruction is not inferred from the running time of a particular algorithm. In contrast, common expected caps admit an exact two-command reduction under a common terminal reward, even when the full catalog has many eligibility classes. Enumerating singletons and pairs solves this subclass in polynomial time without an accuracy parameter. It is distinct from the retained exact common-prefix subclass: neither identical caps nor identical eligibility is imposed by the general model. For general caps, rational price-path bounds support a finite, grid-free branch-and-bound algorithm. Every explored or unexplored leaf carries a valid upper bound, and every incumbent is an original-space lottery policy. The resulting interval is valid even when a run reaches its resource limit. The unrestricted exact search can still be exponential. The improvement is a polynomial node oracle and compact, data-dependent certificates, not an assertion of polynomial exact optimization in the NP-complete regime.

The path representation also strengthens preprocessing. A price path forced to contain a given command bounds every book using that command, including books in which it is the anchor. Comparing this bound with a feasible incumbent gives safe exclusion. This is complementary to the retained, inexpensive non-anchor envelope, which preserves targets uniformly but ignores the accepted promise and budget. Neither

test guarantees a positive removal rate. We therefore measure exclusion rates, threshold slacks, missed exclusions, anchor removals, and total computation rather than identify an unchanged incumbent as proof that a new command is irrelevant.

The closest algorithmic antecedents are Lagrangian constrained-path methods, not a new form of generic branching. Handler and Zang (1980) combine dual bounds with path enumeration, and Beasley and Christofides (1989) use Lagrangian bounds for tree search and problem reduction. Fisher (1981) explains the general integer-programming role of such relaxations. Those mechanisms are used here, not claimed as inventions. The necessary additional theorem is that the uneliminated renewal problem has the stated exact priced arc functions, despite individual caps, pre-draw service, realization ceilings, and endogenous command choices. The proof also supplies a heterogeneous-reward extension that the common-reward primal rearrangement cannot provide.

The retained resource approximation relates to constrained-path discretization (Hassin, 1992; Lorenz and Raz, 2001), capped service allocation to separable convex resource allocation (Patriksson, 2008), and the uniform-grid acceleration to totally monotone matrix search (Aggarwal et al., 1987). Ordered quantizer design (Wu, 1991) and adjacent mean-preserving randomization (Pagès and Wilbertz, 2012) supply related ordering and interpolation ideas. They do not by themselves allocate a fixed aggregate obligation jointly with a paid, limited command book and history-specific realization restrictions. Our novelty claims concern the two exact decompositions, their constructive connection to the original constraints, and the hardness and common-cap tractability results. Gridding, dual weak bounds, and incumbent-based exclusion retain their established interpretations.

The computational study separates theorem validation from solver evaluation. Exact regressions compare against every feasible book on small models and reject altered certificates. A matched-limit study varies histories, catalog size, budget, accuracy, charges, and reward heterogeneity. It includes cases outside the allotted enumeration range, records interrupted runs, and reports proof-verification cost separately. Historical fine-accuracy cases are rerun rather than compared to timings from another machine. Additional zero-width runs expose anytime behavior, and near-threshold screening cases compare the two safe rules with exact inclusion relevance. Every operational input remains synthetic. The paper makes a methodological case; it does not infer field calibration or a universal speed advantage from those experiments.

2 Finite-Catalog Renewal Design

There are k histories. History j has probability $\pi_j > 0$, expected-obligation cap $b_j \in [0, 1]$, and realization ceiling $\tau_j \geq b_j$, with $\sum_j \pi_j = 1$. Write $\bar{b} = \sum_j \pi_j b_j$. Intermediate service has cost $h_j : [0, b_j] \rightarrow \mathbb{R}_+$, continuous, convex and nondecreasing, with $h_j(0) = 0$. The initial common terminal reward $f : [0, 1] \rightarrow \mathbb{R}$ is strictly increasing and concave. The catalog is $\mathcal{A} = \{a_1 < \dots < a_N\} \subset [0, 1]$. Opening command a_i costs $\rho_i \geq 0$.

A nonempty selected book $c = (c_1 < \dots < c_s)$ has $s \leq m$. After observing j , the provider selects

deterministic pre-draw service $y_j \geq 0$ and a lottery C_j supported on c . The problem is

$$V^*(B) = \max_{c, y, C} \left\{ \sum_j \pi_j [\mathbb{E}f(C_j) - h_j(y_j)] - \sum_{z \in c} \rho(z) \right\}, \quad (1)$$

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq \tau_j \quad \text{almost surely}, \quad \sum_j \pi_j t_j = B. \quad (2)$$

Here $B \in [0, \bar{b}]$ is accepted, rather than optimized away. The common-risk specialization is $\tau_j = b_j + \delta$. Expected and realized obligations are different restrictions. The terminal symbol does not carry a free history index, inherited service tier, or shared random seed. The cardinality m measures the command alphabet, not total implementation storage: targets, encoder logic, probability precision, program constants, and certificates are additional resources. The mathematical results below concern the stated finite catalog; continuous-design results have a separate approximation error.

For a feasible book, let $d_j = \max(c \cap [0, \tau_j])$, and let $F_{j,c}$ interpolate f linearly on this eligible prefix. A singleton prefix has the constant value $f(d_j)$ at its only point.

Proposition 2.1 (Canonical response). *A book is feasible if and only if $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$. At target $c_1 \leq t \leq b_j$, its optimal response is*

$$v_{j,c}(t) = F_{j,c}(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (3)$$

It is attained by $y_j = (t - d_j)_+$ and an adjacent eligible lottery with mean $\min(t, d_j)$. The response is concave in t .

Proof. Every feasible command satisfies $C_j \leq \tau_j$ since $y_j \geq 0$. At mean $\mu = \mathbb{E}C_j$, concavity on the ordered support bounds the reward by $F_{j,c}(\mu)$. The payoff is at most $F_{j,c}(\mu) - h_j(t - \mu)$ for $c_1 \leq \mu \leq \min(t, d_j)$. Both terms are nondecreasing in μ , so the upper endpoint maximizes this bound. When $t \leq d_j$, adjacent interpolation has zero intermediate service and support below τ_j . When $t > d_j$, use command d_j surely and service $t - d_j$; the realized sum is $t \leq b_j \leq \tau_j$. Thus the bound is attained without weakening either restriction. Interpolation slopes decrease and are nonnegative; the tail slopes of $-h_j$ are nonpositive and nonincreasing. This proves concavity. Feasible targets fill the entire weighted interval $[c_1, \bar{b}]$, proving necessity and sufficiency. $\square$

The algorithmic specialization is

$$f(x) = rx - \frac{1}{2}qx^2, \quad r \geq q \geq 0, \quad r > 0, \quad h_j(y) = \frac{1}{2}\gamma_j y^2, \quad \gamma_j \geq 0. \quad (4)$$

All probabilities, caps, ceilings, catalog points, shape parameters, opening charges, and requested accuracies are rationally encoded in exact-arithmetic claims. Optimization identities allow real nonnegative charges; bit-complexity assertions do not assign a finite encoding to arbitrary real numbers. A common Lipschitz constant for the canonical responses is

$$L = \max\{r, \max_j \gamma_j\} > 0. \quad (5)$$

Continuity of the cost on its compact domain makes the stated extrema attained; every rational quadratic instance already satisfies this regularity. A fixed-book allocation can be solved by filling positive chord slopes in decreasing order, then zero-cost service capacity, then quadratic service tails. The service breakpoints are $\gamma_j(b_j - d_j)_+$. This takes $O(k s + k \log(k + 1))$ rational operations with sorted marginal events. Alternatively, a direct supporting-price sweep verifies

$$\sum_j \pi_j v_{j,c}(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\}. \quad (6)$$

Eligible knots, caps, and clipped stationary tail points suffice to check each scalar maximum. This certificate proves optimality for that book, not over all books.

3 Constructive Selected-Boundary Decomposition

This section assumes the common reward f in Section 2. For a selected book $c = (u_1 < \dots < u_s)$, write $a = u_1$ and $d_j = \max\{u_i : u_i \leq \tau_j\}$.

Lemma 3.1 (Global terminal-first allocation). *An optimal fixed-book canonical allocation cannot have both $t_i > d_i$ and $t_j < \min(b_j, d_j)$.*

Proof. Choose $0 < \delta \leq \min\{t_i - d_i, \pi_j[\min(b_j, d_j) - t_j]/\pi_i\}$. Reduce t_i by δ and increase t_j by $\pi_i\delta/\pi_j$. The weighted change is zero. The donor's terminal reward is unchanged and its service cost weakly decreases; the recipient's strictly increasing terminal interpolant improves. Both responses remain feasible by Proposition 2.1, contradicting optimality. $\square$

For selected neighbors $u < v$, define

$$\delta_j(u, v) = \mathbf{1}_{\{\tau_j \geq v\}} [\min(b_j, v) - u]_+, \quad T_{uv} = \sum_j \pi_j \delta_j(u, v), \quad (7)$$

$$J_{uv} = \{j : u \leq \tau_j < v\}, \quad Q_{uv} = \sum_{j \in J_{uv}} \pi_j (b_j - u)_+. \quad (8)$$

The capped service value on an exit set is

$$H_{uv}(y) = \min \left\{ \sum_{j \in J_{uv}} \pi_j h_j(z_j) : 0 \leq z_j \leq (b_j - u)_+, \sum_{j \in J_{uv}} \pi_j z_j = y \right\}, \quad 0 \leq y \leq Q_{uv}. \quad (9)$$

Empty sets have zero capacity and zero value at zero. Let $\sigma_{uv} = [f(v) - f(u)]/(v - u)$ and

$$\Phi_{uv}(x) = \sigma_{uv} \min(x, T_{uv}) - H_{uv}((x - T_{uv})_+), \quad 0 \leq x \leq T_{uv} + Q_{uv}. \quad (10)$$

For the terminal edge $(u, \dagger)$, use $T_{u\dagger} = 0$, $J_{u\dagger} = \{j : \tau_j \geq u\}$, and $\Phi_{u\dagger} = -H_{u\dagger}$. The functions Φ are concave. Under the quadratic primitives, they are L -Lipschitz with L as in (5).

Lemma 3.2 (Nested terminal layers). *Start each history at target a . Fill the selected terminal layers in increasing edge order. Once every preceding layer is full, each history with $\delta_j(u, v) > 0$ is at target u . Every weighted amount $x \in [0, T_{uv}]$ in the next layer has a feasible individual allocation. After all terminal layers are full, history j is at $\min(b_j, d_j)$; its remaining service capacity belongs to exactly one exit set.*

Proof. A history with positive increment on (u, v) has $b_j > u$ and $\tau_j \geq v$. It is eligible for all earlier selected commands, and the increments of those earlier layers telescope to $u - a$. This proves the first assertion, including a history whose cap will truncate the current layer.

For any desired x , initialize residual $r = x$ and visit histories in any fixed order. Give history j increment $e_j = \min\{\delta_j(u, v), r/\pi_j\}$ and reduce r by $\pi_j e_j$. Since $x \leq \sum_j \pi_j \delta_j(u, v)$, the final residual is zero. Each affected target lies in $[u, \min(b_j, v)]$ and is implemented by the adjacent lottery on u, v with upper probability $e_j/(v - u)$. It has no intermediate service and both realizations are below τ_j . Histories with zero increments keep their previously feasible responses.

If a layer is partial, stop: its predecessors are full and its successors unused. If every layer is full, the increments for history j telescope to $\min(b_j, d_j) - a$. It exits exactly after d_j , at an ordinary crossing edge or at the terminal edge. Thus its remaining capacity is $(b_j - d_j)_+$ and appears in exactly one H . Any feasible service vector on the exit sets implements target $\min(b_j, d_j) + z_j$. A positive z_j requires $b_j > d_j$, and its implementation is the sure command d_j plus service z_j ; their sum is at most $b_j \leq \tau_j$. Zero-capacity histories require no service. The exit sets are disjoint, so these implementations combine without conflicting allocations. $\square$

Lemma 3.3 (Implementable rearrangement). *For a fixed selected path, any feasible arc-resource vector can be rearranged, with unchanged total resource and no smaller payoff, into full earlier terminal layers, at most one partial terminal layer, and service only after all terminal layers are full. That rearranged payoff is attained by original history-level policies.*

Proof. Split the resource on each arc into its terminal and service parts in (10). If any terminal capacity remains unused while some service is positive, transfer the smaller of the available terminal capacity and a positive service amount to that terminal layer. The terminal gain is positive, while reducing service weakly lowers cost. Repeat until there is no such pair. This operation never changes charges or the sum of resources.

Next, if an earlier terminal layer is unfinished and a later one is positive, transfer the smaller of the unfinished capacity and the later mass. Selected chord slopes are positive and nonincreasing, so payoff cannot decrease. Filling layers successively yields the asserted form after finitely many exhausted donors or filled recipients. Each arc still respects its capacity: terminal mass is at most T_e , and service, when present, is at most Q_e after that arc's terminal part is full. Allocate any remaining service by minimizers of the corresponding H_e ; these minimizers exist by compactness and continuity. Lemma 3.2 implements the resulting terminal pattern and the disjoint service allocations. Its gross payoff is precisely $f(a) + \sum_e \Phi_e(x_e)$ for the rearranged vector. An arbitrary original vector need not be implemented at its unrearranged payoff; the weak improvement is the comparison needed for exact optimal values. $\square$

Theorem 3.4 (Exact selected-boundary resource path). *Let $P(c)$ contain the successive selected edges and the terminal edge of a feasible book. Then*

$$\sum_{e \in P(c)} (T_e + Q_e) = \bar{b} - a, \quad (11)$$

$$V_c(B) = f(a) - \sum_{z \in c} \rho(z) + \max_{\substack{0 \leq x_e \leq T_e + Q_e \\ \sum_e x_e = B - a}} \sum_{e \in P(c)} \Phi_e(x_e). \quad (12)$$

Maximizing over paths with at most m selected commands solves the original joint problem exactly. Reconstruction fills terminal layers in order, uses the greedy partial-layer allocation of Lemma 3.2, and then solves the disjoint capped service allocations. The resulting policy obeys the original promise, expected caps, realization ceilings, and command budget.

Proof. For history j , the terminal increments sum to $\min(b_j, d_j) - a$ and its unique service capacity is $(b_j - d_j)_+$. Their sum is $b_j - a$, proving (11) after weighting. In an original optimum, Lemma 3.1 places service after terminal capacity. Among terminal layers, exchanging equal weighted amounts orders the common decreasing chord slopes; Lemma 3.2 justifies the individual feasibility of that ordering. If all terminal capacity is full, the exit-set cost minimizations describe all remaining feasible service allocations. Hence an original optimum supplies an arc allocation of the same value. Conversely, Lemma 3.3 implements a weakly improved version of every feasible arc allocation. The two comparisons yield equality of the attained maxima. Each selected charge is subtracted once, and the terminal edge does not open an additional command. $\square$

Corollary 3.5 (Endogenous eligibility). *A book of s commands has at most s nonempty selected-prefix classes. Inserting unselected candidates changes neither its arc functions nor its fixed-book optimum.*

Proof. The nonempty exit sets partition histories according to their last eligible selected command. Their definitions and terminal increments depend on the selected endpoints, not intervening unused candidates. $\square$

3.1 An eligibility-independent capacity potential

The path-capacity identity has a stronger local form. It explains why eligibility affects returns, but does not create a hidden resource state.

Proposition 3.6 (Capacity potential). *Define $p(u) = \sum_j \pi_j \min(b_j, u)$ for catalog vertices and $p(\dagger) = \bar{b}$. Every selected-boundary arc satisfies*

$$T_{uv} + Q_{uv} = p(v) - p(u), \quad Q_{u\dagger} = p(\dagger) - p(u). \quad (13)$$

Thus its capacity is independent of the realization ceilings as long as $\tau_j \geq b_j$. For a feasible anchor a , $p(a) = a$.

Proof. Fix one history and an ordinary edge $u < v$. If $\tau_j < u$, then $b_j < u$, so both its contribution to the arc and $\min(b_j, v) - \min(b_j, u)$ are zero. If $u \leq \tau_j < v$, then $b_j < v$ and its exit-service capacity

is $(b_j - u)_+ = \min(b_j, v) - \min(b_j, u)$. If $\tau_j \geq v$, its terminal increment is $[\min(b_j, v) - u]_+$, which equals the same difference. These cases partition the histories. Multiplication by the probabilities and summation prove the first identity. For the terminal edge, $\tau_j < u$ again implies $(b_j - u)_+ = 0$, giving $Q_{u^\dagger} = \sum_j \pi_j (b_j - u)_+ = \bar{b} - p(u)$. Finally, feasibility implies $a \leq b_j$ for every history. $\square$

Changing a ceiling can still change an arc's payoff through its exit set. Proposition 3.6 asserts capacity invariance, not value continuity across a threshold. It also identifies the renewal reduction as a capacity-conservative configuration of the type used in the production model below.

4 A Complexity Frontier for Joint Design

Equal expected caps do not require equal realization ceilings. A dense catalog may therefore give a different full-catalog prefix to every history even in the following exact regime.

Theorem 4.1 (Two commands suffice under common caps). *Assume $b_j = b$ for every history, the common increasing concave reward f , convex nondecreasing service costs, and nonnegative command charges. For any feasible $B \leq b$ and command budget $m \geq 1$, an optimal book exists with at most $\min(m, 2)$ commands. With rational quadratic primitives, enumerating feasible singletons and pairs and solving their fixed-book allocations uses*

$$O(N^2 k \log(k + 1))$$

rational arithmetic operations and polynomial bit complexity, independently of the number of eligibility classes and of any accuracy parameter.

Proof. The assertion is immediate for $m = 1$. Consider any feasible book with at least two commands and an optimal fixed-book allocation in the form of Lemma 3.2.

If the book contains B , the singleton $\{B\}$ is feasible. For any feasible policy, monotonicity and Jensen's inequality give

$$\sum_j \pi_j \{\mathbb{E}f(C_j) - h_j(y_j)\} \leq \sum_j \pi_j f(t_j) \leq f(B).$$

The singleton attains this upper bound at every target B and costs no more than a book containing it. It weakly improves that book.

If the largest selected command is $u < B$, all selected commands are below b and hence eligible for every history. All terminal layers can be filled up to u before service is used; the residual $B - u$ is service. The resulting targets are at least u . Deleting all other commands leaves the same terminal command, service allocation, targets, and gross value, while weakly reducing charges.

In the remaining case, let $u < B < v$ be the selected neighbors bracketing B . Every selected command through u is below b and eligible for every history. Completing the preceding layers puts every target at u . If $v \leq b$, the (u, v) layer has aggregate capacity $v - u$; the residual $B - u$ is smaller. The two-command book therefore implements the unique partial-layer pattern of the larger book, with no later terminal layer or service needed.

If $v > b$, histories admitting v have their cap below that command. They have no terminal increment beyond the (u, v) layer and no positive service capacity after it. Histories not admitting v admit no later selected command, so their last eligible selected command is u . For targets at least u , both groups have exactly the same canonical responses under the full book and under $\{u, v\}$. Later commands have zero useful capacity, and earlier commands have already been fully traversed. Deleting them preserves the optimized gross value. In all cases a singleton or pair weakly improves the original book after charges, proving the reduction.

There are at most $N + \binom{N}{2}$ retained possibilities. A two-command fixed-book response is computed by its history-level marginal events followed by capped convex quadratic service allocation, costing $O(k \log(k+1))$ rational operations. The finite sums, sorted marginal ratios, and water-level equations have polynomial rational encoding lengths. Comparing the resulting rational values proves the arithmetic and bit claims. $\square$

The result is not a claim that every optimal book is small: additional zero-charge commands can remain in an equally good book. It also cannot be extended to arbitrary cap heterogeneity by the same argument. With caps $1/4, 1/2, 3/4$, positive probabilities, $B = \bar{b}$, strictly concave common reward, zero service costs and charges, and a catalog containing these three caps, the gross Jensen upper bound $\sum_j \pi_j f(b_j)$ is attained by all three deterministic commands. Attaining it requires the respective commands; two cannot suffice. Conversely, even under a common cap, two commands can be necessary. With one history, $b = B = 1/4$, $\tau = 1/2$, catalog $\{0, 1/2\}$, zero costs and charges, and $f(x) = 2x - x^2/2$, the pair yields $7/16$ whereas the only feasible singleton yields zero.

The common-cap theorem and the retained common-prefix theorem in the electronic companion are different polynomial subclasses. Common caps allow many eligibility prefixes; common eligibility allows heterogeneous caps. Neither restriction is imposed on the price-path algorithm below. The next result shows why exact polynomial optimization cannot simply be extended to unrestricted heterogeneous caps unless $P = NP$.

4.1 NP-completeness with linear reward and free service

The exact common-cap theorem has a genuine boundary. Heterogeneous caps and paid command activation can encode a discrete target even when the reward and service primitives are linear and zero, respectively.

Theorem 4.2 (Complexity of exact finite-catalog design). *The rational quadratic joint-design decision problem, given a rational threshold ζ and asking whether $V^*(B) \geq \zeta$, is NP-complete. NP-hardness persists with the fixed common reward $f(x) = x$, zero intermediate-service costs, uniform probabilities, $\tau_j = b_j$, and a nonbinding command budget $m = N$.*

Proof. Membership in NP follows by supplying a selected book. Exact rational fixed-book allocation is computable in polynomial time and bit complexity, as in Proposition 5.2; its rational value can be compared with ζ . This argument applies to the full rational quadratic model, not to arbitrary unencoded convex functions.

For hardness, start with positive integer SUBSET SUM weights $w_1, \dots, w_n, n \geq 2$, total $A = \sum_i w_i$, and target $1 \leq W \leq A$. The NP-completeness of this binary-encoded problem is classical (Karp, 1972). Put

$$K = n^2 A, \quad \ell_i = \frac{i-1}{n}, \quad z_i = \ell_i + \frac{w_i}{nA}, \quad \pi_i = \frac{1}{n}, \quad b_i = \tau_i = z_i. \quad (14)$$

The catalog consists of the $2n$ distinct commands ℓ_i, z_i . Their order is $\ell_i < z_i < \ell_{i+1}$ for $i < n$, since positive weights and $n \geq 2$ imply $w_i < A$; also $z_n < 1$. Set $m = 2n$, charge zero for every ℓ_i , and charge $w_i/(2K)$ for z_i . Use $f(x) = x$ and $h_i = 0$. Define

$$D_0 = \frac{1}{n} \sum_i \ell_i, \quad B = D_0 + \frac{W}{K}, \quad \zeta = D_0 + \frac{W}{2K}. \quad (15)$$

These rational inputs have polynomial binary length. The root promise is feasible because $\bar{b} = D_0 + A/K$ and the minimum catalog command is zero.

Every feasible book can be augmented by all free commands ℓ_i without increasing its charge or exceeding the nonbinding budget. Enlarging its lottery support cannot reduce its attainable value. Hence an optimal book contains all those commands. For an optional subset S of commands z_i , history i has last eligible command z_i when $i \in S$ and ℓ_i otherwise; higher free commands are above its realization ceiling. Its aggregate terminal capacity is

$$D(S) = D_0 + \frac{\sum_{i \in S} w_i}{K}.$$

Let $s = \sum_{i \in S} w_i$. With linear unit reward and free service, the optimized gross payoff is $\min\{B, D(S)\}$. For $B \leq D(S)$, lotteries between zero and each last eligible command provide every aggregate terminal mean up to $D(S)$. For $B > D(S)$, fill those terminal commands and use the remaining feasible service capacity $\bar{b} - D(S)$. Thus both parts of this gross-value formula are attained in the original constraints. Subtracting the optional charges gives the exact identity

$$V_S(B) = \min\{B, D_0 + s/K\} - \frac{s}{2K} = \zeta - \frac{|s - W|}{2K}. \quad (16)$$

Consequently $V^*(B) \geq \zeta$ holds if and only if a subset sums exactly to W . This is a polynomial reduction and proves NP-hardness under all the stated restrictions. $\square$

The reduction does not prove strong NP-hardness, hardness at a fixed small command budget, or a parameterized lower bound. It does establish that unrestricted exact polynomial-time optimization would imply $P = NP$. It is consistent with the additive scheme: a no-instance can be only $1/(2K)$ below the threshold, even though $f(x) = x$ has Lipschitz constant one. Resolving that distinction may require an inverse accuracy proportional to the numerical integer total. This identifies an encoding-sensitive barrier, not a proof that every instance requires the uniform grid or that its exact $1/\varepsilon$ dependence is optimal. The common-cap and common-prefix algorithms remain exact polynomial exceptions.

5 Exact Price Paths with Heterogeneous Rewards

We now replace the common reward by a strictly increasing concave function f_j for each history. All other constraints and the common ordered catalog are unchanged. Proposition 2.1 holds history by history with interpolant $F_{j,c}$ of f_j . In particular,

$$v_{j,c}(t) = F_{j,c}(\min(t, d_j)) - h_j((t - d_j)_+)$$

is concave on the feasible target interval. The primal rearrangement in Section 3 need not hold across histories with differently ordered rewards. The following priced identity does not use that rearrangement.

For price $\lambda \in \mathbb{R}$, let

$$\sigma_{uv}^j = \frac{f_j(v) - f_j(u)}{v - u}, \quad K_j(u; \lambda) = \max_{0 \leq y \leq (b_j - u)_+} \{-h_j(y) - \lambda y\}, \quad (17)$$

$$\begin{aligned} A_{uv}(\lambda) &= \sum_{j: \tau_j \geq v} \pi_j [\min(b_j, v) - u]_+ (\sigma_{uv}^j - \lambda)_+ \\ &\quad + \sum_{j: u \leq \tau_j < v} \pi_j K_j(u; \lambda), \end{aligned} \quad (18)$$

$$A_{u^\dagger}(\lambda) = \sum_{j: \tau_j \geq u} \pi_j K_j(u; \lambda). \quad (19)$$

These supports are finite. At a feasible anchor a , define

$$Z_a(\lambda) = \sum_j \pi_j f_j(a) - \lambda a - \rho(a).$$

Theorem 5.1 (Exact priced selected-boundary identity). *For every fixed feasible book $c = (a = u_1 < \dots < u_s)$,*

$$\begin{aligned} &\sum_j \pi_j \max_{a \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\} - \sum_{z \in c} \rho(z) \\ &= Z_a(\lambda) + \sum_{i=1}^{s-1} [A_{u_i u_{i+1}}(\lambda) - \rho(u_{i+1})] + A_{u_s^\dagger}(\lambda). \end{aligned} \quad (20)$$

Consequently, a longest ordered path with at most m selected vertices computes the exact maximum of the left side over all feasible books. Adding λB gives an upper bound on the original joint optimum, including under heterogeneous terminal rewards.

Proof. Fix a history and its eligible selected prefix. Beginning at a , its terminal interpolation consists of successive segments of lengths $[\min(b_j, v) - u]_+$, followed, when $b_j > d_j$, by its convex service-cost tail. The terminal chord slopes are positive and nonincreasing *within this history*. Subtracting λt therefore selects exactly all positive marginal terminal segments, with arbitrary selection on zero-slope ties. Their gain is the sum of segment length times $(\sigma_{uv}^j - \lambda)_+$.

For $\lambda \geq 0$, avoidable service cannot improve the priced objective, so its support is zero. For $\lambda < 0$, every

positive terminal chord exceeds the price and is fully traversed before a service tail is selected. Maximizing the remaining concave tail gives $K_j(d_j; \lambda)$. Thus the complete scalar support is

$$f_j(a) - \lambda a + \sum_{\substack{(u,v) \text{ selected} \\ v \leq \tau_j}} [\min(b_j, v) - u]_+ (\sigma_{uv}^j - \lambda)_+ + K_j(d_j; \lambda).$$

The terminal terms are owned by their selected edges, and the last term by the unique exit edge or terminal edge of that history. Weighted summation gives (20), with each command charge subtracted once. The identity needs no comparison between chord slopes of different histories.

For an original feasible target vector, $\sum_j \pi_j t_j = B$ cancels the subtracted price when λB is added. Taking each scalar maximum and then maximizing the book proves weak duality. Every feasible book has a feasible anchor, so the ordered-path maximization omits none. $\square$

5.1 One recurrence for all anchors and restricted books

Let $\alpha = \min(B, \min_j b_j)$. At a fixed price, initialize

$$D_1(v) = Z_{a_v}(\lambda) \quad \text{when } a_v \leq \alpha, \quad D_1(v) = -\infty \quad \text{otherwise.}$$

For $\ell = 2, \dots, m$,

$$D_\ell(v) = \max_{u < v} \{D_{\ell-1}(u) + A_{a_u a_v}(\lambda) - \rho_v\}, \quad U(\lambda) = \lambda B + \max_{\ell, v} \{D_\ell(v) + A_{a_v \dagger}(\lambda)\}. \quad (21)$$

Unlike the resource-grid recurrence, this recurrence handles every anchor simultaneously. A predecessor pointer supplies a maximizing book. Its targets are recovered at the original promise by exact fixed-book allocation, not by adopting its generally infeasible priced target mean.

The same oracle permits a mandatory vertex set M and a forbidden set F . Remove F , disallow starting after the first mandatory vertex, crossing over a mandatory vertex, or terminating before the last mandatory vertex. If $|M| > m$ or $M \cap F \neq \emptyset$, the family is empty. Otherwise the allowed paths are exactly the books that contain M and avoid F : each mandatory vertex is either visited or would have to be skipped at the start, at an edge, or at the end. Prefix counts test these conditions in constant time per edge. Denote the resulting upper bound by $U_{M,F}(\lambda)$.

Proposition 5.2 (Rational quadratic price oracle). *Suppose $f_j(x) = r_j x - q_j x^2/2$ with rational $r_j \geq q_j \geq 0$, $r_j > 0$, and $h_j(y) = \gamma_j y^2/2$ with rational $\gamma_j \geq 0$. For a rational price, a restricted price path and its upper bound are computed in $O(kN^2 + mN^2)$ arithmetic operations after the input is read, with polynomial bit complexity. Exact fixed-book recovery costs $O(ks \log(ks + 1) + k \log(k + 1))$ operations for s selected commands.*

Proof. For $\gamma_j > 0$, the maximizing service is $\min\{(b_j - u)_+, \max(0, -\lambda/\gamma_j)\}$. When $\gamma_j = 0$, use zero service for nonnegative price and the full cap for negative price. Every support is a rational quadratic evaluation. Computing all edge sums directly costs $O(kN^2)$; the recurrence uses mN^2 comparisons and additions. It is

unaffected by the number of distinct full-catalog eligibility prefixes.

For recovery, sort the at most ks history-level terminal marginal segments by decreasing slope, preserving each history's internal order on ties. Fill them until the promise is met or all terminal capacity is used. Fill zero-cost service next and then solve the remaining capped quadratic water-level equation. This constructs an original feasible policy. Concavity of the individual responses supplies its supporting price, proving fixed-book optimality. All sorting keys and values are rational combinations of polynomially many input numbers. Exact common-denominator scaling for path comparisons and the resulting sums have polynomial bit lengths. The number of chosen search prices is not part of this single-oracle bound. $\square$

For a common quadratic reward, event-prefix sums accelerate repeated evaluations: $T_{uv} = p(v) - p(u) - Q_{uv}$ and the terminal term is $T_{uv}(\sigma_{uv} - \lambda)_+$. After sorting exit events for each predecessor, one price sweep accumulates service supports in $O(kN + N^2)$ operations before the $O(mN^2)$ path calculation. The reference implementation uses this optimization, exact integer-scaled comparisons, and a bounded cache of price kernels. Heterogeneous rewards use the direct edge sums. Arithmetic counts, rational bit complexity, and measured interpreter time are distinct quantities.

A small price gap does not imply strong duality over books. For one history, cap and ceiling one, $f(x) = x$, zero service cost, catalog $\{0, 1/2\}$, charges $0, 1/10$, and budget two, the optimized value is $\max\{0, \min(B, 1/2) - 1/10\}$. Its slope increases at $B = 1/10$. At $B = 1/20$ the true optimum is zero but its concave dual majorant is positive. The next section closes such gaps by a complete discrete partition rather than silently treating a single price as a global optimality certificate.

5.2 A heterogeneous-reward primal extension

The priced identity also suggests the correct generalization of the primal arc payoff. Simply replacing the common reward by different rewards in (10) would be incorrect, because one chord slope no longer describes all histories. Instead, define the within-edge terminal allocation value

$$R_{uv}(x) = \max \left\{ \sum_{j: \tau_j \geq v} \pi_j \sigma_{uv}^j e_j : 0 \leq e_j \leq \delta_j(u, v), \sum_j \pi_j e_j = x \right\}, \quad 0 \leq x \leq T_{uv}. \quad (22)$$

It is continuous, increasing and concave, with marginal slopes drawn from the history-specific chord slopes. Set

$$\tilde{\Phi}_{uv}(x) = R_{uv}(\min(x, T_{uv})) - H_{uv}((x - T_{uv})_+). \quad (23)$$

The terminal edge still has payoff $-H_{u+}$. All terminal slopes are positive and all service marginal returns are nonpositive, so this joined payoff is concave.

Theorem 5.3 (History-level component packing). *For heterogeneous increasing concave terminal rewards, replace $f(a)$ in (12) by $\sum_j \pi_j f_j(a)$ and every Φ by $\tilde{\Phi}$. The resulting scalar-resource path identity is exact. Its capacities remain those in (11). For any feasible arc-resource vector, within-edge optimizers followed by history-level packing produce a feasible original policy with the same individual component totals and no smaller payoff.*

Proof. For a fixed book, introduce a component $e_{j,uv} \in [0, \delta_j(u, v)]$ for every eligible terminal segment, and service $z_j \in [0, (b_j - d_j)_+]$. Temporarily drop the requirement that a history's earlier components must be filled before its later ones. The relaxed objective, apart from charges, is

$$\sum_j \pi_j \left[f_j(a) + \sum_{(u,v)} \sigma_{uv}^j e_{j,uv} - h_j(z_j) \right], \quad \sum_j \pi_j \left[\sum_{(u,v)} e_{j,uv} + z_j \right] = B - a.$$

Every original canonical policy embeds in this relaxation by filling its eligible segments in order and then its service tail. Conversely, fix any relaxed component allocation and define $t_j = a + \sum_{(u,v)} e_{j,uv} + z_j$. The sum of that history's component capacities is exactly $b_j - a$, so $a \leq t_j \leq b_j$. Its components can be packed in their original order while preserving t_j : move service into any unfinished positive terminal segment, then move mass from a later positive segment into any unfinished earlier segment. Each move weakly improves payoff, because the history's terminal slopes are decreasing and positive and service cost is nondecreasing. This comparison is entirely within a history; it needs no common ordering across histories. At termination the components implement its canonical response at t_j . Proposition 2.1 supplies the adjacent lottery or sure-command service policy, with every realization constraint intact. The weighted promise is unchanged by packing. Thus relaxation and original problem have the same optimum.

Now regroup the relaxed components by selected edge, assigning z_j to the unique exit edge of history j . Groups share only the scalar total-resource equality; their individual component bounds are otherwise disjoint. For a fixed group resource x , optimal allocation fills positive terminal returns first, whose optimum is R_{uv} , and then minimizes the exit-set service cost H_{uv} . This is exactly (23). Maximizing over group resources is therefore the same relaxation already proved exact. The component-capacity telescoping is unchanged. Within-edge optimizers followed by the preceding history-level packing provide the asserted constructive recovery. $\square$

In the quadratic specialization, R_{uv} is evaluated by sorting the eligible history slopes and filling weighted segment capacities. Its Lipschitz constant is at most $\max_j r_j$. The service part is bounded by $\max_j \gamma_j$. Hence the polynomial additive resource construction also extends to heterogeneous rewards with these new kernels. The retained uniform-grid implementation used in the comparison study has only the common-reward kernels; it is not represented as an executed heterogeneous solver. The new price-path implementation and independent certificates do support heterogeneous rewards. This separates the broader theorem from the exact scope of the benchmarked software.

6 Finite Exact Search and Original-Instance Certificates

6.1 Price bounds, feasible policies, and discrete coverage

For a restricted family (M, F) , every evaluated price yields an upper bound $U_{M,F}(\lambda)$. Recovering the price-maximizing book at B gives a feasible lower value. Neither the priced mean nor a solver's completion status is a feasible-policy witness. Maintain a global incumbent $\underline{V}$ and, for each live leaf, its best completed price bound. The universal bound $\sum_j \pi_j f_j(1)$ initializes a leaf before its first price evaluation.

The reference search evaluates a bounded number of prices at each node. The target range of a maximizing book gives a valid subgradient direction for the convex price bound. A bracket starts at $-\max_j \gamma_j$ and $\max_j r_j$; candidate prices also include the exact supporting price of the recovered fixed-book policy. Bisection supplies a distinct rational price when a candidate repeats. Bracketing is a search heuristic, not a premise for validity: every completed price produces a valid upper bound independently of how it was chosen.

If a node bound is at most $\underline{V} + \varepsilon$, it need not be split. Otherwise select an undecided candidate and create the two children in which it is respectively required and forbidden. Both inherit the parent's bound until a sharper completed calculation is available. Mandatory and forbidden vertices are enforced by the recurrence in Section 5. A family with one possible book is evaluated at that book's exact supporting price. Every feasible original book remains in exactly one leaf family.

Theorem 6.1 (Grid-free finite exact algorithm). *Under the rational quadratic assumptions of Proposition 5.2, the preceding algorithm returns, after every completed node or price calculation, an original feasible policy and a rational interval*

$$\underline{V} \leq V^*(B) \leq \bar{V}, \quad \bar{V} = \max\{\underline{V}, U_{M,F} : (M, F) \text{ is a current leaf}\}. \quad (24)$$

An empty leaf is omitted. With no resource limit and zero requested width, finite binary branching and exact singleton-family evaluation terminate at $\underline{V} = \bar{V}$. The full binary cover has at most $2^{N+1} - 1$ nodes. Each completed price oracle has the polynomial complexity of Proposition 5.2. At positive tolerance, or under interruption, the actual width $\bar{V} - \underline{V}$ is a valid instance-dependent additive error bound; it has no mandatory inverse-tolerance resource grid.

Proof. The initial incumbent is an original feasible singleton at the minimum command. Every subsequent incumbent is recovered at B and respects the same constraints. Theorem 5.1 bounds every book in a leaf. Requiring or forbidding a previously undecided candidate partitions its family exactly, so inherited bounds remain valid on both children. A node with no allowed path is empty; all nonempty leaf families have feasible target allocations by Proposition 2.1. These facts prove (24), including leaves not yet processed or calculations interrupted before completion.

Along a root-to-leaf chain, a candidate is decided at most once. After at most N splits the family is empty or fixes one book. Concave fixed-book allocation has a supporting price at B , and its scalar maximizers attain the price bound at that same promise. Consequently the restricted priced optimum at that price equals the recovered value. A fixed finite number of price attempts per nonterminal node followed by an undecided split cannot prevent this finite termination. The elementary depth- N binary-tree count supplies the worst-case bound. Rational formulas, finite supports, and original-space policies establish exact arithmetic throughout. $\square$

The worst-case exponential bound is deliberate. Theorem 4.2 proves NP-completeness of the unrestricted decision problem. The present algorithm supplies a finite exact method with a polynomial node relaxation and a data-dependent certified stopping rule; it does not assert a polynomial worst-case exact running time. The exact common-cap and common-prefix theorems give polynomial subcases. For either reward model

with the appropriate arc kernels, the uniform-resource scheme in Section 7 retains a separate worst-case polynomial additive guarantee. A valid combined interval can take the maximum of the feasible lower values and the minimum of the price-tree and resource-grid upper bounds. No assumption of strong duality across books is required.

6.2 Independent verification and proof economics

A price certificate contains the original model, its catalog and charges, a feasible lottery policy, and the complete binary coverage tree. A priced leaf records its rational price and upper bound; a still-unprocessed leaf may retain the universal bound. The independent checker reconstructs arc supports by direct endpoint and stationary-point maximization, then uses a backward at-most-budget recurrence rather than the optimizer’s forward exact-length recurrence. It checks every branch, every mandatory or forbidden restriction, and every leaf bound. A parent price bound may be looser than a child’s recomputed bound; the required comparison is dominance, not equality.

The lower witness retains nonnegative service, lottery masses, command support, every positive-probability realization ceiling, all expected caps, and the exact weighted promise. The checker subtracts the original selected charges and recomputes the global interval. A separately supplied instance digest can bind the proof to the requested input; a self-reported digest alone is not authentication. The new checker raises explicit errors and does not rely on assertions being enabled. The inherited grid checker continues to require ordinary Python execution, as documented in its retained source.

The proof record stores $O(k + s + N + P)$ rational fields and tree entries up to the history-level lottery representation, where P is the number of retained nodes. It need not serialize the scalar-resource tables. Verification recomputes polynomial price paths at the leaf prices; it is not free, and a large branching tree can still be expensive. Input and price bit lengths contribute to actual byte counts. The study reports compressed and uncompressed size, checking time, peak process memory, and resource-limit outcomes separately from the command alphabet.

6.3 Incumbent-based exclusion, including anchors

The uniform anchor-envelope rule retained in the electronic companion cannot delete a possible anchor. The priced oracle provides a complementary test.

Proposition 6.2 (Forced-command price exclusion). *Let v_0 be the value of an original feasible incumbent. For any candidate z , including a possible anchor, let $U_z(\lambda)$ be the price-path upper bound with z mandatory. If $U_z(\lambda) < v_0$, no optimal book contains z . All commands certified by such strict inequalities may be removed simultaneously without changing the optimal value. Under a weak inequality, simultaneous deletion is safe only with an independently retained incumbent avoiding all deleted commands.*

Proof. Every book containing z lies in its forced family and has value at most $U_z(\lambda)$. A strict comparison excludes it from optimality and also implies that the incumbent does not contain it. Therefore every optimal book and the incumbent avoid the union of strictly excluded candidates. The reduced problem retains an optimal policy and introduces no new one. For weak inequalities, any book lost by a deletion has value at

most v_0 , while an incumbent avoiding the entire deletion set attains v_0 in the retained problem. This proves that qualified version as well. $\square$

This is the familiar incumbent-based Lagrangian reduction principle (Beasley and Christofides, 1989; Fisher, 1981), applied through an exact original-instance price oracle. Its benefit here is that the bound incorporates the accepted promise, command budget, and other selected alternatives. No uniform removal fraction follows. Reusing the arc supports for one price, evaluating every forced candidate costs $O(mN^3)$ additional arithmetic operations. The cheap $O(kN)$ non-anchor envelope remains a different option with an order-independent target-box guarantee. The screening study reports both costs and compares their exclusions with exact inclusion relevance rather than declaring one rule universally preferable.

7 A Polynomial Additive Fallback

For the common-reward model, or the heterogeneous kernels of Theorem 5.3, the selected-boundary representation yields a complementary worst-case guarantee. An arc has concave L -Lipschitz payoff and capacity $T_e + Q_e$. Every book with feasible anchor a has capacity $\bar{b} - a \geq B - a$. Round each of its at most m arc resources down to multiples of η , including the terminal edge. Accept total resource in $[B - a - m\eta, B - a]$. Every accepted path has enough spare capacity to restore its missing resource without changing its selected commands.

Theorem 7.1 (Original-promise additive fallback). *For rational increasing concave quadratic rewards and quadratic service costs, let $L = \max_j \{r_j, \gamma_j\}$ and set $\eta = \varepsilon / (2Lm)$ and let S_η be the best accepted grid-path value over all feasible anchors. Exact fixed-book recovery at B yields*

$$\underline{V} \leq V^*(B) \leq S_\eta + Lm\eta, \quad S_\eta + Lm\eta - \underline{V} \leq \varepsilon.$$

With $Q = \lceil 2Lm/\varepsilon \rceil + 1$, direct kernel construction and resource convolution have arithmetic bound

$$O(kN^2Q \log(k+1) + mN^3Q \log(Q+1)), \quad (25)$$

and bit complexity polynomial in the numerical grid size and binary input length.

Proof. Rounding the resources of an optimal path loses at most $Lm\eta$ in payoff. Conversely, repairing a maximizing accepted path changes its coordinates upward by total at most $m\eta$ and loses at most $Lm\eta$. The constructive identity implements the repaired vector with no smaller payoff; exact fixed-book recovery can improve it further. This proves the interval. Precompute each of $O(N^2Q)$ arc values by sorted terminal components and capped quadratic service allocation. At a fixed edge, a concave grid kernel gives an anti-Monge max-plus convolution even if the previous Bellman values are not concave. Monotone row search costs $O(Q \log(Q+1))$ per transition. Counting anchors and command lengths gives (25). The full recurrence, rational scaling argument, and independent checker are retained in the electronic companion. $\square$

Writing normalized accuracy as $\delta = \varepsilon/L$ exposes the effective terms

$$O\left(\frac{kmN^2}{\delta} \log(k+1) + \frac{m^2N^3}{\delta} \log(m/\delta+2)\right).$$

Thus polynomial arithmetic does not by itself imply practical scalability. At a prescribed absolute error the dependence is on L/ε , not its binary encoding length. Signed net payoffs and values close to zero preclude a general multiplicative interpretation. The price-tree method instead stops at its observed original-instance width, sometimes at exact equality, but has an exponential worst-case discrete tree. These are complementary guarantees, not a claim that the uniform $1/\varepsilon$ dependence is necessary. The broader heterogeneous-reward guarantee uses the newly derived within-edge profiles, not an unchanged common-reward kernel. The inherited software used as the uniform baseline remains restricted to common rewards.

8 Computational Evaluation

8.1 Protocol and limits

The fixed study contains 30 cases and 142 applicable method runs. One-factor sweeps vary histories through 768, catalog size through 65, command budget through 12, absolute accuracy through 10^{-6} , and opening-charge scale. Four cases have heterogeneous terminal rewards; six are the original difficult historical models. Caps, probabilities, service curvatures, and ceilings are heterogeneous. Repeated base cases in different sweeps are intentional paired comparisons, not independent statistical observations. Table 1 reports every new price-path case; Table 2 gives the paired historical reruns.

Each method executes in a fresh process with one thread, a four-second algorithm-time alarm and a 2-GiB address-space cap. Imports are timed separately; peak resident memory is measured rather than assumed equal. The mixed-integer reference uses the original support-indicator formulation with 32-segment tangent and secant service envelopes and a two-second solver limit for each call. Total construction and both solves remain subject to the common alarm. A native solver can delay delivery of a Python signal; a separate five-second algorithm-phase process limit records that outcome. This distinction, actual elapsed time, and every limit event are retained.

The baselines are the inherited uniform-resource grid, exact book enumeration, class-mean box subdivision, and the direct numerical mixed-integer model. The inherited grid and box implementations support common rewards only and are not run on the four heterogeneous-reward cases. The price method makes at most 12 price attempts per node and evaluates at most 255 nodes. Rational proof checking is separate, with a 60-second verification alarm. An interrupted old routine is not assigned its unfinished computations as a proof: only an independently feasible incumbent and a universal upper bound survive unless the routine has returned a checkable certificate. Numerical mixed-integer status is never interpreted as rational certification. The full limits and input hashes were committed before these runs.

8.2 Scaling, accuracy, and unsuccessful cases

Across the original study, price paths produce 10 exact closures, 19 positive-width tolerance completions, and 1 resource-limited outcomes. All 30 resulting price certificates pass independent verification. Table 3 retains every method’s outcome. The larger-catalog cases demonstrate performance beyond what the allotted exhaustive enumeration can complete; they do not establish polynomial worst-case exact search. The largest history case illustrates the remaining limit. Its enclosing interval is reported even when it misses the requested tolerance.

The historical comparison is no longer between newly timed code and a different publication environment. All five applicable methods are rerun under the same caps. Enumeration remains attractive on these small catalogs, and numerical mixed-integer solutions can be fast. The benefit of the new path oracle is a substantially smaller exact-rational proof workload than the uniform resource grid on these cases, not a universal ordering of solver times. Compressed price certificates in the full study range from 541 to 5895 bytes. Table 1 also reports checking time and peak process memory; small proof files do not imply zero verification cost.

Ten separate zero-width runs expose the search’s anytime behavior in Table 4. They include all six historical models and the largest two catalog and budget configurations. Of these, 10 close exactly within the stated limits. Every per-oracle and per-node checkpoint remains in the execution record. Unlike a uniform grid, a smaller requested tolerance does not necessarily require more resource states: a supporting price or a completed discrete partition can close an interval exactly. The construction’s worst-case guarantee remains finite exponential search, as stated in Theorem 6.1.

After deriving the complexity theorem, we separately froze four SUBSET SUM challenge configurations before executing them. Their outcomes occupy the lower panel of Table 3. These are deliberately difficult structural cases, not representative operational inputs. The source integer weights are even and the target odd; integer subset-sum dynamic programming supplies their exact reference distance. The new method also encounters node and time limits on this family. These failures are informative: the price bound does not remove the discrete obstruction proved in Theorem 4.2. Grid and box comparisons use width $1/(8K)$, while price and enumeration request exact closure, as explicitly declared in the addendum; their success counts are not pooled as identical-accuracy outcomes.

8.3 Screening and independent correctness checks

Table 5 reports all 24 screening cases, crossing eight nonuniform-cap models and charge multipliers 0.9, 1, 1.1 around the target-box envelope. The catalogs include several possible low anchors and a mixture of retained and deleted commands; promises and budgets vary. Exact inclusion and exclusion optima identify useful, tied, and strictly irrelevant candidates. Forced price bounds remove 45 possible-anchor occurrences across these cases. Rates, missed strictly irrelevant commands, wholly ineligible removals, and every charge slack are recorded. The stronger test incurs incumbent-search and bound-checking costs; no generic preprocessing speedup or removal guarantee is inferred. Three additional sharpness configurations make the same command necessary below the exclusion threshold, tied at equality, and excluded above it.

The regression suite verifies 130 models, including 60 common-cap models and 30 heterogeneous-reward models. All 130 global price intervals close at the exhaustive optimum. Independent singleton/pair certificates pass on the 60 common-cap cases, 40 of which have more than one eligibility class. The suite checks 2,644 fixed books, 390 restricted-price oracles, 2,035 nested-layer reconstructions, and the heterogeneous component construction and optimal component equality on all 2,644 books. A deliberate positive root-duality-gap example requires branching; a one-node interruption retains a valid incomplete-width certificate. Nine altered certificates are rejected. The separately declared hardness validation checks all 1692 optional subsets on 60 reductions and all 3004 feasible catalog books in the smaller reductions. Every distance identity and yes/no equivalence agrees. These finite checks verify implementation consistency; the universal statements rest on the proofs.

9 Conclusion

Finite-catalog joint renewal design has an exact path structure and a nontrivial complexity boundary. The original constraints yield constructive selected-boundary resources, and history-level component packing extends that structure to heterogeneous terminal rewards. Exact priced paths turn the same history-level supports into polynomial node bounds. Common expected caps admit an exact two-command reduction, whereas heterogeneous caps and paid activation already give NP-completeness with unit linear reward, zero service costs, and a nonbinding command budget.

These results support complementary algorithms rather than one universal solver claim. Uniform resource gridding gives a polynomial additive guarantee with explicit numerical accuracy dependence. Grid-free price branching gives compact, independently checked original-instance intervals and finite exact termination, with a potentially exponential tree. The experiments expose both favorable larger-catalog cases and difficult reduction instances. Promise-dependent forced-command bounds extend safe exclusion to anchors but need not repay their cost on every model.

The mathematical model still uses one aggregate obligation, a common ordered catalog with prefix eligibility, additive opening charges and pre-draw service. The heterogeneous extension does not establish a multiple-resource or nonprefix result. All reported operational inputs are synthetic; no field calibration or operational tolerance value is inferred. The theory, constructive recovery, and reproducible comparisons supply the methodological contribution.

Code and Data Accessibility

The accompanying `CODE_AND_DATA.zip` contains the current sources, rational inputs, frozen protocols, execution records, exact certificates, numerical comparisons, and reproduction commands, together with unchanged baseline dependencies. Historical computational records remain at their original revision paths. Complete older readers remain in the repository; the current root-reader predecessors are also in the archive. The companion supplies retained proofs and diagnostics. The README explains execution, verification requirements, and the preservation and build manifests.

References

- Aggarwal A, Klawe MM, Moran S, Shor P, Wilber R (1987) Geometric applications of a matrix-searching algorithm. *Algorithmica* 2:195–208. doi:10.1007/BF01840359.
- Beasley JE, Christofides N (1989) An algorithm for the resource constrained shortest path problem. *Networks* 19(4):379–394. doi:10.1002/net.3230190402.
- Fisher ML (1981) The Lagrangian relaxation method for solving integer programming problems. *Management Science* 27(1):1–18. doi:10.1287/mnsc.27.1.1.
- Handler GY, Zang I (1980) A dual algorithm for the constrained shortest path problem. *Networks* 10(4):293–309. doi:10.1002/net.3230100403.
- Hassin R (1992) Approximation schemes for the restricted shortest path problem. *Mathematics of Operations Research* 17(1):36–42. doi:10.1287/moor.17.1.36.
- Karp RM (1972) Reducibility among combinatorial problems. Miller RE, Thatcher JW, eds. *Complexity of Computer Computations* (Plenum Press, New York), 85–103. doi:10.1007/978-1-4684-2001-2_9.
- Lorenz DH, Raz D (2001) A simple efficient approximation scheme for the restricted shortest path problem. *Operations Research Letters* 28(5):213–219. doi:10.1016/S0167-6377(01)00069-4.
- Pagès G, Wilbertz B (2012) Intrinsic stationarity for vector quantization: Foundation of dual quantization. Author manuscript, arXiv:1010.4642v2, March 27.
- Patriksson M (2008) A survey on the continuous nonlinear resource allocation problem. *European Journal of Operational Research* 185(1):1–46. doi:10.1016/j.ejor.2006.12.006.
- Virtanen P, Gommers R, Oliphant TE, et al. (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272. doi:10.1038/s41592-019-0686-2.
- Wu X (1991) Optimal quantization by matrix searching. *Journal of Algorithms* 12(4):663–673. doi:10.1016/0196-6774(91)90039-2.
- Zhang H (2012) Solving an infinite horizon adverse selection model through finite policy graphs. *Operations Research* 60(4):850–864. doi:10.1287/opre.1120.1056.

Table 1: Arbitrary-eligibility scaling under the common four-second algorithm budget.

Case	k	N	m	E	Seconds	Width	Bytes	Check (ms)	Memory
Histories 12	12	25	4	9	0.043	1.3e-04	657	14.40	27.0
Histories 48	48	25	4	16	0.268	9.5e-04	994	37.47	27.0
Histories 192	192	25	4	21	0.877	7.0e-04	2106	132.43	27.0
Histories 768	768	25	4	21	4.011*	0.00252	5895	508.52	27.0
Catalog 9	48	9	4	7	1.403	0.00100	1297	111.13	27.0
Catalog 17	48	17	4	12	2.412	1.8e-04	1288	126.55	27.0
Catalog 33	48	33	4	21	0.252	2.0e-04	1056	49.40	27.1
Catalog 65	48	65	4	28	0.635	5.3e-04	1171	109.86	27.1
Budget 2	48	25	2	16	0.856	4.3e-04	1190	131.34	27.1
Budget 4	48	25	4	16	0.264	9.5e-04	994	36.96	27.1
Budget 8	48	25	8	16	0.269	4.7e-04	995	39.83	27.1
Budget 12	48	25	12	16	0.274	4.7e-04	997	43.50	27.1
Accuracy 1/100	48	25	4	16	0.164	0.00839	992	37.11	27.1
Accuracy 1/1000	48	25	4	16	0.273	9.5e-04	994	39.90	27.1
Accuracy 1/10000	48	25	4	16	0.582	0	1076	101.05	27.1
Accuracy 1/1000000	48	25	4	16	0.576	0	1076	99.71	27.1
Charges 0	48	25	4	16	0.368	4.9e-04	989	36.64	27.1
Charges 1/1000	48	25	4	16	0.262	9.5e-04	994	36.43	27.1
Charges 1/100	48	25	4	16	0.130	0	951	37.15	27.1
Charges 1/10	48	25	4	16	0.127	0	969	36.77	27.1
Different rewards	12	9	4	7	0.011	0	604	8.38	27.1
Different rewards	12	17	4	9	0.026	0	650	22.34	27.1
Different rewards	24	9	4	7	0.021	0	733	15.86	27.1
Different rewards	24	17	4	11	0.051	4.2e-04	791	43.38	27.1

Widths and bytes refer to independently checked rational certificates. Zero width denotes exact closure. An asterisk marks a time or node limit; it does not invalidate the interval. Memory is peak resident process memory in MiB under a common 2-GiB address-space cap. Check is additional verification time. Except in the named accuracy sweep, requested width is 10^{-3} . Charges and accuracy vary on the same base model; these repeated anchors of the one-factor sweeps are not independent samples. Heterogeneous rows correspond to $(k, N) = (12, 9), (12, 17), (24, 9), (24, 17)$.

Table 2: Paired reruns of the six historical fine-accuracy cases.

Seed	Price seconds	Price width	Enumeration	Uniform grid	Class boxes	Mixed integer
49500	0.049	0	0.021	4.004 limit	4.003 limit	0.044
49501	0.138	2.7e-04	0.023	4.004 limit	4.003 limit	0.038
49502	0.006	5.2e-04	0.023	4.003 limit	4.003 limit	0.020
49503	0.154	0	0.017	3.562	4.002 limit	0.041
49504	0.006	3.2e-05	0.023	4.003 limit	2.117	0.020
49505	0.109	0	0.022	4.003 limit	4.003 limit	0.026

Times are measured seconds under the same four-second algorithm allowance and 2-GiB address-space cap. The mixed-integer column includes construction and both envelopes, each with a two-second solver limit; it is not an envelope-error allowance. Numerical mixed-integer bounds are not rational certificates. Complete primal/dual values, solver nodes, model sizes, residuals, envelope errors, and actual per-envelope solver times appear in the companion and raw records. Interrupted old solvers retain only their separately valid fallback interval, not an uncertified intermediate result.

Table 3: All matched-limit outcomes, including unsuccessful runs.

Method	Runs	Exact	Tolerance	Numerical	Resource limit
Price paths	30	10	19	0	1
Uniform grid	26	0	1	0	25
Book enumeration	30	12	0	0	18
Class boxes	26	0	3	0	23
Mixed integer	30	0	0	16	14

Histories	Commands	Price	Uniform	Enumeration	Class boxes	Mixed integer
6	12	Exact	Time limit	Exact	Time limit	Numerical
10	20	Time limit	Time limit	Time limit	Time limit	Numerical
14	28	Time limit	Time limit	Time limit	Time limit	Time limit
18	36	Time limit	Time limit	Time limit	Time limit	Numerical

Top panel: the original 30-case study, 142 method runs. The inherited grid and class-box implementations do not accept heterogeneous rewards, so each has 26 applicable cases. Bottom panel: all four separately preregistered SUBSET SUM reduction challenges, not an operational sample. Price and enumeration request exact closure; grid and class-box width is $1/(8K)$. Constructed-case integer subset-sum dynamic programming provides exact reference values independently of these methods. Resource limits remain outcomes, not discarded cases. A numerical outcome only means that both numerical envelope calls returned; individual solver statuses and widths remain reported.

Table 4: Time-to-gap records for the separately requested zero-width runs.

Case	Width at 0.1 s	Width at 0.5 s	Width at 1 s	Width at 4 s	Final width	Seconds
catalog-33	0.12917	1.4e-04	1.1e-04	0	0	1.111
catalog-65	–	0.01354	0	0	0	0.793
budget-8	0.00789	2.9e-04	0	0	0	0.797
budget-12	0.00789	2.9e-04	0	0	0	0.803
historical-49500	0	0	0	0	0	0.049
historical-49501	0.00803	0	0	0	0	0.147
historical-49502	0	0	0	0	0	0.009
historical-49503	0.00148	0	0	0	0	0.155
historical-49504	0	0	0	0	0	0.009
historical-49505	0.00283	0	0	0	0	0.107

Each entry uses the latest completed price or node calculation at or before the indicated time. A dash means that no completed price calculation had yet been recorded; the initialized universal bound still exists. Final bounds are independently checked. Once a run terminates its final trace continues to describe the incumbent interval. These zero-width reruns are separate from the fixed-tolerance study and use the same resource caps. Full per-oracle traces, not just these checkpoints, are retained.

Table 5: Screening rates and costs on all 24 near-threshold, mixed-catalog cases.

Rule	Minimum	Lower quartile	Median	Upper quartile	Maximum	Anchors
Target-box envelope	0	0.125	0.250	0.250	0.375	0
Forced price paths	0.250	0.375	0.625	0.625	0.750	45

Calculation	Median seconds	Minimum seconds	Maximum seconds	Description
Envelope	3.2e-04	2.8e-04	3.5e-04	Bound evaluation
Incumbent	0.0226	0.0068	0.0840	Price search
Forced paths	0.0368	0.0348	0.0424	Bounds and checks
Exact reference	0.0241	0.0063	0.0595	All feasible books

Rates are fractions of eight catalog commands removed. Anchor counts aggregate all cases, not distinct physical commands. Forced tests include all possible anchors and use feasible-incumbent price information, not an exact-reference value supplied for free. The exact reference is only used for evaluation. A stronger forced test costs more and can miss truly irrelevant commands; it is not advertised as universally preferable preprocessing. The companion reports wholly ineligible removals, charge slacks, missed exclusions, all target-command triples, and sharpness neighbors in which the same command is useful, tied, or excluded.